SILVER OAK UNIVERSITY

EDUCATION TO INNOVATION

KHETICONNECT

Smart Agritech Direct Farmer-to-Consumer & Real-Time Crop Bidding

Project Code - COT-66

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Purpose

KhetiConnect is a smart agritech platform that connects farmers, wholesale buyers, and consumers through a transparent and direct digital marketplace. It provides real-time mandi rates, live crop bidding, retail shopping, and a community portal so that farmers get better prices, buyers get verified produce, and everyone benefits from a fair and efficient agricultural ecosystem.

Abstract

KhetiConnect is a web-based agritech platform that enables direct trade between farmers, buyers, and consumers. It offers real-time mandi rates, live crop bidding, retail shopping with cart and secure checkout, and a community portal for knowledge sharing. The platform reduces dependence on middlemen, ensures price transparency, and builds trust through verified profiles and transparent transactions. It is designed to be simple, fast, and accessible for users across rural and urban areas.

Literature Survey

e-NAM provides market access and price discovery but has complex onboarding and a poor user experience for farmers.

Agribazaar and DeHaat support large-scale B2B trade but are rigid and not suitable for dynamic pricing.

Traditional e-grocery platforms focus on urban consumers, offer low farm-gate prices, and do not connect buyers directly with farmers.

KhetiConnect combines real-time mandi analytics, live bidding, direct retail sales, and a community portal in one platform, solving the limitations of existing systems.

Methodology

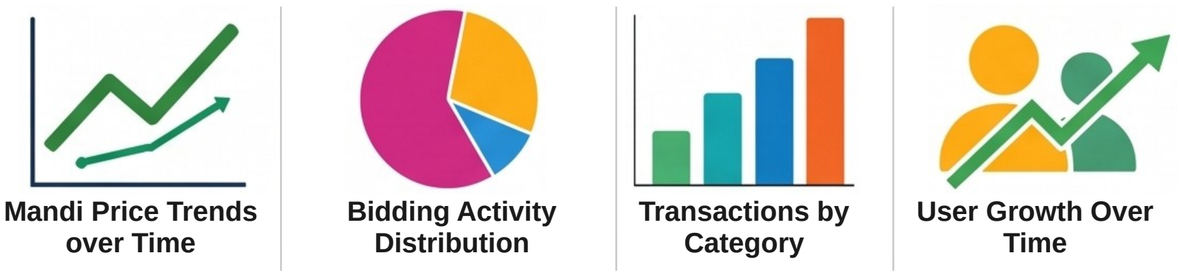
We use a user-centered and iterative development approach. First, we study user needs and define the requirements for farmers, buyers, and consumers. Then we design the system architecture and database, followed by frontend and backend implementation. After that, we integrate features like the mandi ticker, bidding engine, cart, and community portal. Finally, we test the platform, fix issues, and improve usability based on feedback.



This process ensures that KhetiConnect is reliable, scalable, and easy to use.

Data Analysis (Data / Tables / Graphs / Images)

We collect real-time mandi rates, bidding activity, and transaction data to monitor platform performance and user trends.



These insights help us understand market behavior, user engagement, and platform efficiency, which guide future improvements.

Results & Discussion

KhetiConnect successfully demonstrates how technology can remove middlemen, bring price transparency, and create a trusted environment for agricultural trade. The live mandi ticker and dynamic bidding system ensure fair price discovery, while the retail cart and secure checkout make buying simple. The community portal adds value through knowledge sharing and trust building among farmers and buyers.

Testing shows that the platform is responsive, easy to use, and effective for both farmers and buyers. Future improvements will focus on stronger verification, real-time notifications, and wider deployment with live mandi data integration.

Conclusion & Future Scope

KhetiConnect makes agricultural trade fair, transparent, and simple by connecting farmers and buyers directly. It improves price discovery, reduces middlemen, and builds trust through verified profiles and real-time data.

In the future, we plan to integrate verified mandi data, secure real payments, strengthen profile verification, add real-time notifications, and deploy the platform for a larger number of users.

References (IEEE Format)

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[2] A. Goyal, "Information, direct access to farmers, and rural market performance in central India," *Amer. Econ. J.: Appl. Econ.*, vol. 2, no. 3, pp. 22–45, 2010.

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